Practical No. 4

Aim: Programming UI Elements (TextView, EditText, Button, ImageButton, CheckBox, RadioGroup, RadioButton, ToggleButton).

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView android:layout_height="match_parent"
   android:layout_width="match_parent"
   xmlns:android="http://schemas.android.com/apk/res/android" >

   <LinearLayout
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:orientation="vertical">

      <EditText
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:id="@+id/edtName"
         android:hint="Enter Name"
         android:textSize="20dp"/>

      <RadioGroup
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:id="@+id/rg">
         <RadioButton
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/rbFemale"
            android:text="Female"
            android:textSize="20dp"/>

         <RadioButton
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/rbMale"
            android:text="Male"
            android:textSize="20dp"/>

         <RadioButton
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/rbOther"
            android:text="Other"
            android:textSize="20dp"/>
      </RadioGroup>
```

```
<CheckBox
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:id="@+id/chbDance"
   android:text="Dancing"
   android:textSize="20dp"/>

<CheckBox
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:id="@+id/chbSing"
   android:text="Singing"
   android:textSize="20dp"/>

<CheckBox
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:id="@+id/chbRead"
   android:text="Reading"
   android:textSize="20dp"/>

<Button
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:id="@+id/btnOk"
   android:text="OK"
   android:textSize="20dp"/>

<TextView
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:id="@+id/tvName"
   android:textSize="20dp"/>

<TextView
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:id="@+id/tvGender"
   android:textSize="20dp"/>

<TextView
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:id="@+id/tvHobbies"
   android:textSize="20dp"/>

<ImageButton
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:id="@+id/imgBtn"
   android:src="@mipmap/ic_launcher_round"/>
```

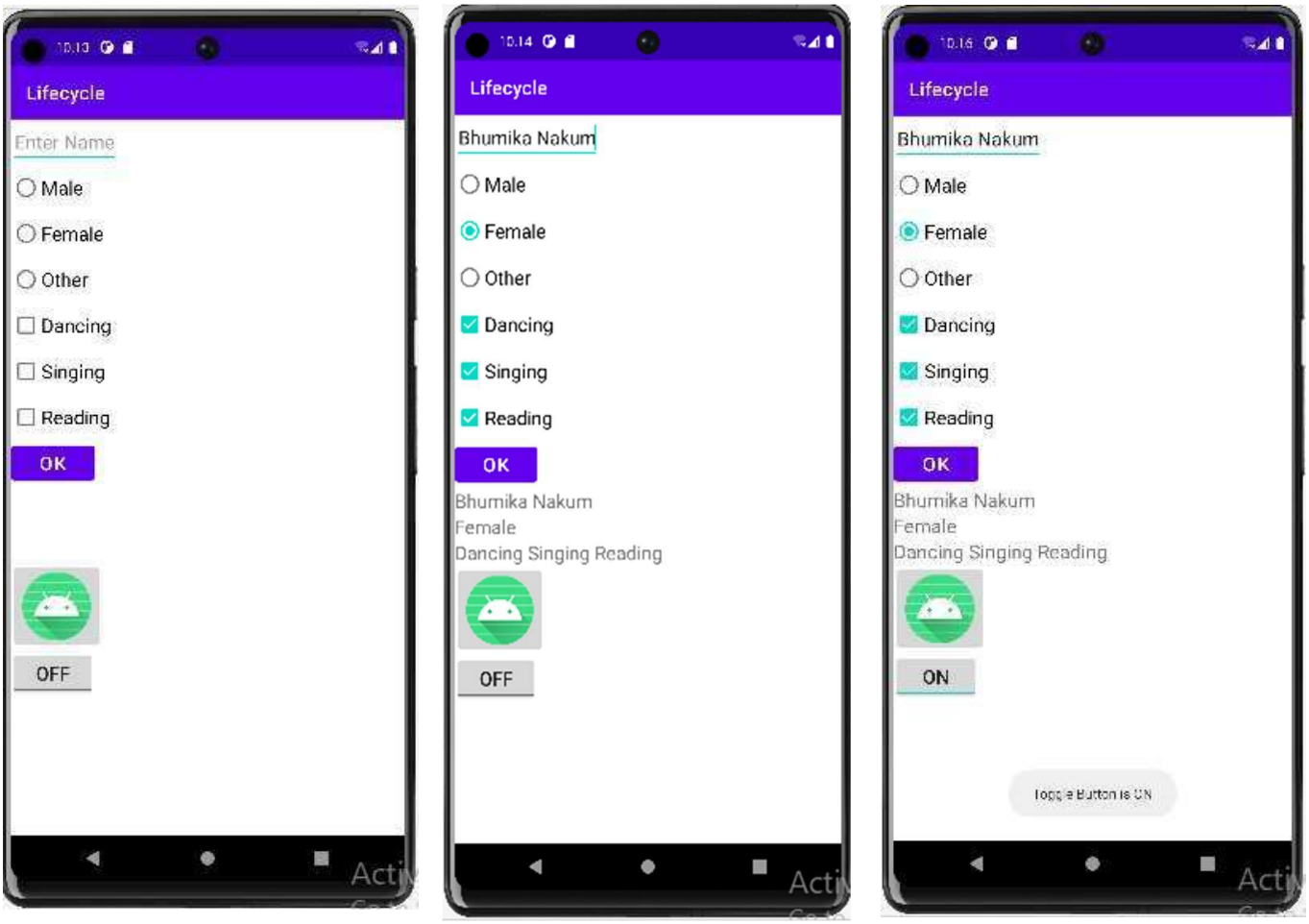
```
        <ToggleButton
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/tg"
            android:textSize="20dp"/>

    </LinearLayout>
</ScrollView>
```

MainActivity.kt

```
package com.example.lifecycle

import android.annotation.SuppressLint
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.util.Log
import android.widget.*

class MainActivity : AppCompatActivity() {

    @SuppressLint("MissingInflatedId")
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val edtName = findViewById<EditText>(R.id.edtName)
        val rg = findViewById<RadioGroup>(R.id.rg)
        val rbMale = findViewById<RadioButton>(R.id.rbMale)
        val rbFemale = findViewById<RadioButton>(R.id.rbFemale)
        val rbOther = findViewById<RadioButton>(R.id.rbOther)
        val chbDance = findViewById<CheckBox>(R.id.chbDance)
        val chbSing = findViewById<CheckBox>(R.id.chbSing)
        val chbRead = findViewById<CheckBox>(R.id.chbRead)
        val btnOk = findViewById<Button>(R.id.btnOk)
        val tvName = findViewById<TextView>(R.id.tvName)
        val tvGender = findViewById<TextView>(R.id.tvGender)
        val tvHobbies = findViewById<TextView>(R.id.tvHobbies)
        val imgBtn = findViewById<ImageButton>(R.id.imgBtn)
        val tg = findViewById<ToggleButton>(R.id.tg)
        var s = ""

        btnOk.setOnClickListener {
            tvName.text = edtName.text

            if(rbMale.isChecked)
                tvGender.text = rbMale.text
            else if (rbFemale.isChecked)
                tvGender.text = rbFemale.text
            else
                tvGender.text = rbOther.text
```

```
        if(chbDance.isChecked)
            s = chbDance.text as String
        if(chbSing.isChecked)
            s = s + " " + chbSing.text as String
        if(chbRead.isChecked)
            s = s + " " + chbRead.text as String
        tvHobbies.text = s
    }

    imgBtn.setOnClickListener {
        edtName.setText("")
        rbMale.isChecked = false
        rbFemale.isChecked = false
        rbOther.isChecked = false
        chbDance.isChecked = false
        chbSing.isChecked = false
        chbRead.isChecked = false
        tvName.setText("")
        tvGender.setText("")
        tvHobbies.setText("")
    }

    tg.setOnCheckedChangeListener { _, isChecked ->
        if(isChecked)
        {
            Toast.makeText(this, "Toggle Button is ON", Toast.LENGTH_LONG).show()
        }
        else
        {
            Toast.makeText(this, "Toggle Button is OFF", Toast.LENGTH_LONG).show()
        }
    }
  }
}
```

Output


Lifecycle
Enter Name
Male
Female
Other
Dancing
Singing
Reading
OK
OFF
Lifecycle
Bhumika Nakum
Male
Female
Other
Dancing
Singing
Reading
OK
Bhumika Nakum
Female
Dancing Singing Reading
OFF
Lifecycle
Bhumika Nakum
Male
Female
Other
Dancing
Singing
Reading
OK
Bhumika Nakum
Female
Dancing Singing Reading
ON